

NLGS-SLAM: Robust Gaussian Splatting SLAM against Non-Lambertian Interference

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Abstract—High-fidelity reconstruction of heterogeneous scenes with unknown camera poses is a crucial yet challenging problem in VR and AR applications. The recent integration of 3D Gaussian splatting with SLAM enables both spatial localization and high-fidelity scene reconstruction. However, widespread Non-Lambertian surfaces in heterogeneous scenes make multi-view geometry-based localization challenging, while their anisotropic reflectance is difficult to represent with low-order spherical harmonics, thereby degrading rendering fidelity. This paper proposes a novel NLGS-SLAM, which aims to achieve high-fidelity reconstruction and real-time rendering for heterogeneous scenes containing Non-Lambertian surfaces. The key insight of our NLGS-SLAM lies in a unified strategy that integrates localization, geometric reconstruction, and Non-Lambertian appearance optimization. This joint strategy not only ensures accurate camera pose estimation and structural consistency, but also effectively models complex surface reflectance, thereby enabling both high-fidelity reconstruction and photorealistic rendering in real time. Specifically, we design an uncertainty-driven detection module for robust segmentation of Non-Lambertian regions, and a neural network predicts 3D Gaussian parameters in a single forward pass. Finally, a Non-Lambertian enhanced scene representation with ray-tracing mechanism is proposed for reconstructing heterogeneous scenes while supporting real-time rendering. Experimental results on public datasets demonstrate that the proposed NLGS-SLAM achieves better reconstruction and significantly improves the rendering fidelity of Non-Lambertian surfaces compared to previous methods. To the best of our knowledge, NLGS-SLAM is one of the first 3D Gaussian splatting SLAM methods aiming at heterogeneous scenes containing Non-Lambertian surfaces, enabling real-time high-fidelity rendering and achieving state-of-the-art reconstruction.

Index Terms—Gaussian Splatting, SLAM, Non-Lambertian, Efficient Rendering.

I. INTRODUCTION

VISUAL simultaneous localization and mapping (SLAM) aims to extract features from temporally continuous images to jointly estimate camera poses and construct maps, forming the basis for 3D scene reconstruction and virtual-real fusion in virtual reality (VR) and augmented reality (AR) [1]–[4]. Over the past decade, SLAM has achieved substantial progress across semantic SLAM [5], dynamic SLAM [6], and scene reconstruction [7], with consistent improvements in

accuracy, robustness, and functionality [8]. However, applying SLAM in heterogeneous real-world scenes remains challenging, as Non-Lambertian surfaces such as mirrors, glass, and specular highlights often degrade robustness and may even cause tracking failures [9]–[11].

The emergence of Neural radiance fields (NeRF) [12] has enabled SLAM-based reconstruction methods to achieve high-fidelity rendering from arbitrary viewpoints, offering many advantages for NeRF-based SLAM in photorealistic rendering and compact scene representation [13], [14]. However, since NeRF suffers from high rendering latency, those NeRF-based SLAM approaches still have limited practical deployment in VR and AR applications. Although subsequent NeRF variants [15]–[17] can significantly reduce rendering latency, the per-pixel ray-tracing still incurs considerable delay.

Recently, 3D Gaussian splatting (3DGS) [18] has demonstrated superior rendering fidelity and substantial speed advantages over NeRF, meeting the real-time rendering requirements of VR and AR. Subsequent work integrating 3D Gaussian splatting with SLAM has consistently enabled accurate simultaneous localization and high-fidelity scene reconstruction [7], [19]–[24]. Some methods have further demonstrated capabilities in semantic reconstruction and robustness to dynamic object interference [25]. However, in heterogeneous scenes, Non-Lambertian surfaces remain a major challenge for both robust localization and high-fidelity reconstruction. Although some methods proposed effective *offline* reconstruction for Non-Lambertian scenes [26]–[28], there are still few works focus on online SLAM scenarios with joint optimization for localization and scene-level 3DGS reconstruction. When facing Non-Lambertian surfaces, previous 3DGS-based SLAM approaches would easily lead to inaccurate localization and unsatisfied scene reconstruction, while pronounced blurring of Non-Lambertian rendering may substantially degrade user immersion, ultimately constraining the broader adoption of VR and AR in complex heterogeneous environments.

In this paper, we propose a novel 3D Gaussian splatting based SLAM, called NLGS-SLAM, which aims to accurately reconstruct heterogeneous scenes containing prominent Non-Lambertian surfaces while achieving real-time and high-fidelity scene rendering. Unlike previous 3D Gaussian splatting SLAM methods that represent the entire scene using low-order spherical harmonics within the splatting rasterization [7], [19]–[24], we propose a new Non-Lambertian enhanced scene representation with ray-tracing mechanism for high-fidelity reconstruction. This approach enhances the anisotropic color

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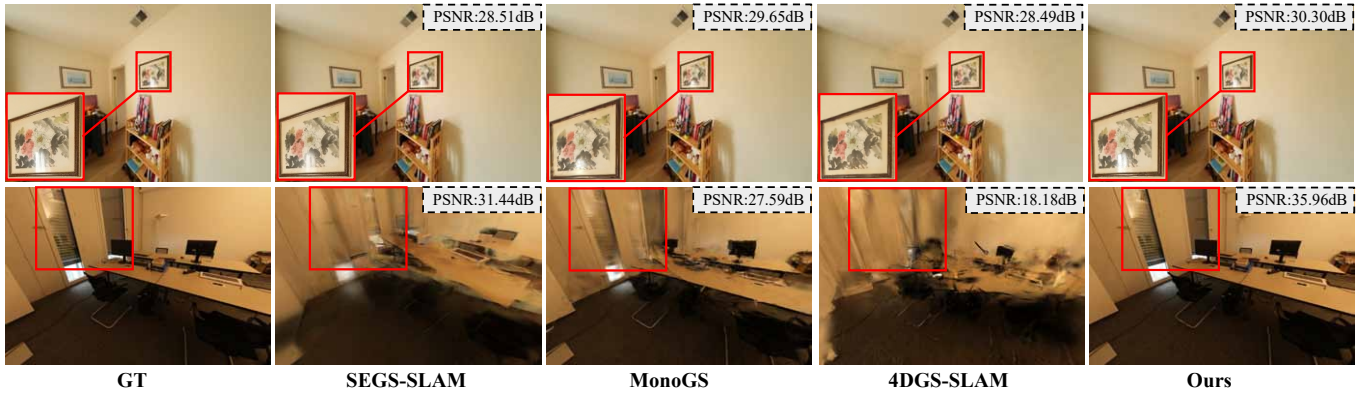


Fig. 1. This paper proposes a novel Gaussian splatting SLAM, i.e., NLGS-SLAM, which aims at the high-fidelity reconstruction of heterogeneous scenes containing Non-Lambertian surfaces. Here we demonstrate final scenes rendering from ScanNet++ [29] cases, with prominent Non-Lambertian regions highlighted. The first row illustrates the results of novel view synthesis, while the second row shows renderings from training views.

representation of these surfaces in a more physically accurate manner, while preserving efficient rendering elsewhere. Moreover, we further design a neural network for direct estimation of 3D Gaussian parameters from image and depth information in a single forward pass. Besides, we segment prominent Non-Lambertian surfaces in 3D scenes using uncertainty-driven detection. Benefiting from the above components, our NLGS-SLAM achieves heterogeneous scene reconstruction with Non-Lambertian surfaces, ensuring both high-fidelity and real-time rendering performance.

As shown in Figure 1, NLGS-SLAM is evaluated on challenging sequences from the ScanNet++ dataset [29]. These sequences feature mirrors, glass, and highlights, with some containing multiple prominent Non-Lambertian surfaces. To demonstrate the general applicability of our method, we further evaluate it on the widely used TUM dataset [30]. We compare NLGS-SLAM with state-of-the-art scene reconstruction SLAM methods, such as MonoGS [19], 4DGS-SLAM [25], and SEGS-SLAM [21]. The experimental results indicate that NLGS-SLAM outperforms these methods in both accuracy and rendering quality, while maintaining real-time rendering performance. To the best of our knowledge, NLGS-SLAM is one of the first 3D Gaussian splatting SLAM methods for heterogeneous scenes with Non-Lambertian surfaces, enabling real-time high-fidelity rendering and achieving state-of-the-art reconstruction.

II. RELATED WORK

A. Scene Reconstruction SLAM

Unlike traditional SLAM methods that struggle to represent fine-grained scene geometry and appearance, iMAP [31] is the first to integrate SLAM with NeRF [12], enabling the joint optimization of a complete 3D map and camera poses. However, its simple MLP architecture struggles to efficiently store rich scene information, which limits its applicability to large-scale reconstruction. To address this issue, NICE-SLAM [14] and Vox-Fusion [32] adopt voxel-based representations for implicit features, enabling more faithful large-scale scene rendering. To address the demand for efficient training and real-time rendering, subsequent approaches have introduced

more structured representations of implicit information, such as octrees [33], [34], hash tables [15], [35], tri-planes [16], and point clouds [17]. Some of these methods even achieve real-time tracking and reconstruction [34], [35], demonstrating great potential for deployment in VR and AR applications. Moreover, the incorporation of loop closure [36] and sub-mapping strategies [37] has significantly improved performance in long-sequence reconstruction. In addition, functional extensions such as semantic mapping [38], multi-model integration [39], and dynamic scene reconstruction [40], [41] have also been explored within the NeRF-based SLAM framework, further broadening its applicability to complex real-world scenarios.

Although NeRF [12] and its variants [15]–[17] demonstrate strong performance in high-fidelity scene reconstruction, 3D Gaussian splatting methods still exhibit significant advantages in efficient training, real-time rendering, and high-quality rendering [18]. In addition, to meet the demand for accurate scene reconstruction, methods such as 2DGS [42], which uses flattened shapes as primitives, and GOF [43], which is based on ray tracing, have also been explored. Efficient integration of 3D Gaussian splatting with SLAM, as demonstrated by MonoGS [19], SplatTAM [7], and Photo-SLAM [23], enables more faithful reconstruction and faster rendering. Moreover, RTG-SLAM [22] uses compact Gaussians with forgetting-preventing optimization, enabling high-quality reconstruction and reduced memory consumption; CG-SLAM [20] selects more valuable Gaussian primitives using an uncertainty model to improve tracking efficiency and accuracy, while GS-ICP SLAM [24] achieves extremely high system speed by sharing the covariances of G-ICP and 3DGS with scale alignment techniques. SEGS-SLAM [21] further enhances scene structure utilization by initializing anchor points from ORB-SLAM [44] point clouds, resulting in significant rendering improvements. Beyond these, Gaussian splatting SLAM methods targeting functional requirements such as dynamic object handling [25] and semantic segmentation [45] have also shown promising performance. In contrast, our work aims to address the interference of prominent Non-Lambertian surfaces on online reconstruction while achieving high-fidelity reconstruction.

B. Non-Lambertian Reconstruction

High-fidelity reconstruction of Non-Lambertian surfaces remains an important and challenging problem in computer graphics [46]. Ref-NeRF [47] estimates surface normals to compute ray reflection directions and enhances the representation of reflection information by encoding these directions. SpecNeRF [48] models prominent light sources in the scene using Gaussians. Although it achieves high-fidelity rendering of specular highlights, the method is difficult to generalize to color anisotropy arising from non-highlight reflections. Building on these approaches, recent developments in 3D Gaussian splatting have further advanced the capabilities of Non-Lambertian surface reconstruction. Existing intrinsic decomposition methods based on 3D Gaussians, such as GS-ID [49], GS-IR [50], and Relightable 3D Gaussians [51], can partially disentangle the complex coupling between lighting and material, but they struggle to generalize to prominent Non-Lambertian surfaces. Moreover, while the use of learnable cubemap [27], [52], [53] or the introduction of dedicated primitives for encoding specular reflections [54], [55] can enhance the richness of specular representation, these approaches are not consistent with the physical principles of light transport. As a result, they fall short of providing an accurate characterization of reflection information. In addition, although Mirror-NeRF [28] and Mirror-3DGS [56] demonstrate impressive reconstruction results for mirrors, their applicability is limited to representing mirror-like specular reflections.

In contrast, EnvGS [26] leverages ray tracing to more physically accurately represent the color anisotropy of Non-Lambertian surfaces using environmental Gaussians. This paradigm jointly optimizes Gaussian rasterization and ray tracing, achieving strong performance in reconstructing Non-Lambertian surfaces, and is easily applicable to a variety of Non-Lambertian surfaces. However, performing ray tracing for all materials in a scene introduces significant time overhead, limiting its application in VR and AR.

C. Generalizable Scene Reconstruction

Traditional scene reconstruction methods typically require multiple iterations for each input image to optimize the parameters of the reconstructed scene, which incurs significant computational and time costs. In contrast, research on generalizable scene reconstruction leverages pre-trained neural networks to encode spatial priors, enabling a single-pass estimation of the 3D scene structure from image information and greatly reducing reconstruction overhead [57], [58]. NeuralRecon [59] employs an offline-trained GRU-based deep neural network to achieve real-time TSDF estimation from image inputs, while GP-Recon [60] further extends generalizable scene reconstruction to NeRF. With the advent of 3D Gaussian splatting, methods such as MVSGaussian [61], FreeSplat [62], and LONG-LRM [63] have enabled scene reconstruction using 3D Gaussians as primitives directly from images. However, these methods generally do not effectively incorporate depth priors, which may constrain their reconstruction performance.

III. METHOD

Given a sequence of temporally continuous RGB-D images $\{I, D\}$ in scenes containing Non-Lambertian surfaces, our NLGS-SLAM aims to simultaneously estimate the poses T_{CW} of the input images and reconstruct a high-fidelity scene $\mathcal{S}$, represented by both basic Gaussian and reflective Gaussians, in an online manner. Specifically, our NLGS-SLAM contains three main modules, including tracking $\mathcal{T}$, prominent Non-Lambertian segmentation $\mathcal{P}$, and efficient mapping with Non-Lambertian enhanced scene representation $\mathcal{M}$. Our key insight is to integrate Lambertian property obtained from $\mathcal{P}$ into $\mathcal{T}$ and $\mathcal{M}$, to formulate a new scene representation enabling high-fidelity, real-time rendering. The pipeline of our NLGS-SLAM is shown in Figure 2.

A. Tracking

We propose a Non-Lambertian enhanced scene representation, i.e., NL-ESR, enabling high-fidelity, real-time rendering. NL-ESR is constructed using both basic and reflective Gaussians, among them, the basic Gaussian shares similar attributes with the vanilla 3D Gaussian [18], but extends a Lambertian attribute lb , and a specular strength attribute $stren$, representing the likelihood of being Lambertian and the intensity of specular reflection, respectively. The reflective Gaussian shares similar attributes with the vanilla 2D Gaussian [42]. With this representation, our NLGS-SLAM adopts a frame-to-model scheme to estimate the camera pose. Specifically, we optimize the camera pose by minimizing both photometric and geometric losses, which are computed by comparing the image rendered by basic Gaussian based on the current camera pose with the ground truth. Unlike previous methods, we introduce the image-space mask of prominent Non-Lambertian surfaces, M_{inl} , into the pose optimization to enable robust tracking against interference from these surfaces. The corresponding photometric loss L_{pho} and geometric loss L_{geo} formulations are given as follows:

$$L_{pho} = E \left[\left\| I(\mathcal{G}, T_{CW})(\sim M_{inl}) - \bar{I}(\sim M_{inl}) \right\|_1 \right], \quad (1)$$

where $I(\mathcal{G}, T_{CW})$ renders the Gaussians $\mathcal{G}$ from T_{CW} , $\bar{I}$ is an observed image, and $\sim M_{inl}$ denotes the complement of M_{inl} .

$$L_{geo} = E \left[\left\| D(\mathcal{G}, T_{CW})(\sim M_{inl}) - \bar{D}(\sim M_{inl}) \right\|_1 \right], \quad (2)$$

where $D(\mathcal{G}, T_{CW})$ is depth rasterization and $\bar{D}$ is the observed depth.

In our system, an initial optimization of the camera pose for the current frame is first performed using photometric and geometric losses without M_{inl} . If the current frame is selected as a keyframe, an uncertainty-driven method is employed to segment prominent Non-Lambertian surfaces in the 3D scene, subsequently obtaining the M_{inl} for the current frame. Once the M_{inl} is available, the scene representation and camera pose are jointly optimized within a local window, with photometric loss Equation (1) and geometric loss Equation (2). For keyframe selection and local window maintenance, a strategy similar to MonoGS [19] is adopted, utilizing criteria such

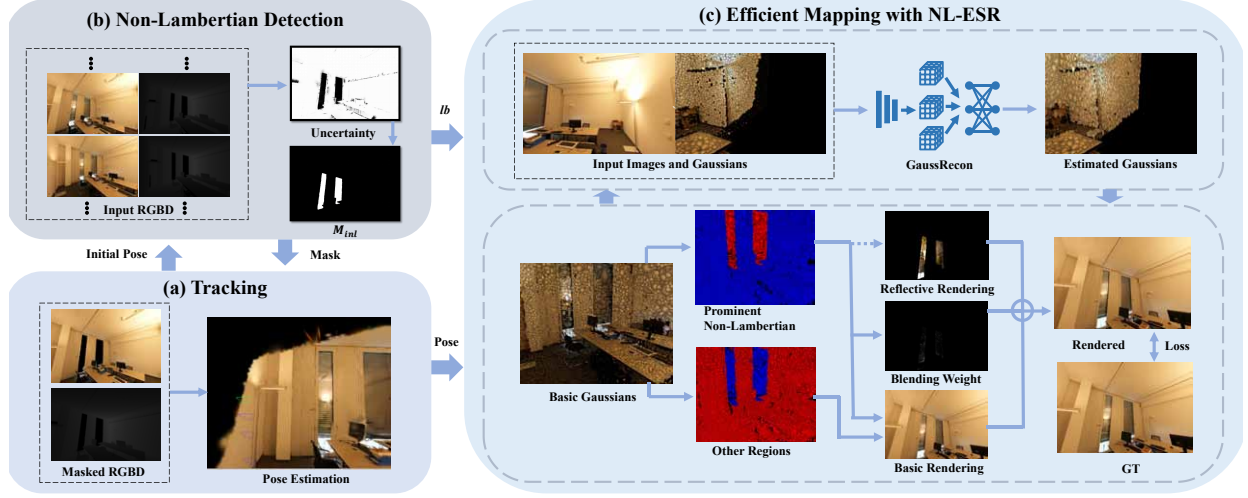


Fig. 2. The pipeline of NLGS-SLAM. Given a sequence of temporally continuous RGB-D images in scenes containing Non-Lambertian surfaces, our NLGS-SLAM aims to simultaneously estimate the poses T_{CW} of the input images and reconstruct a high-fidelity scene, represented by both basic and reflective Gaussians, in an online manner. We propose a frame-to-model pose estimation method that effectively incorporates Lambertian property, enabling robust online camera pose estimation (Section III-A). The prominent Non-Lambertian surfaces in 3D scenes are segmented using uncertainty-driven detection (Section III-B). Furthermore, a new representation for high-fidelity scene reconstruction is proposed. It employs a joint optimization of basic and reflective Gaussians, enabling real-time, high-fidelity scene rendering. In addition, we design a neural network that directly estimates 3D Gaussian parameters from image and depth information in a single forward pass (Section III-C).

as the co-visibility ratio of 3D Gaussians and the translation distance between two frames.

B. Prominent Non-Lambertian Detection

The detection of prominent Non-Lambertian surfaces has long been an important yet challenging problem in computer vision and computer graphics [11]. Existing neural network-based segmentation approaches either suffer from poor generalization ability or focus only on a limited set of prominent Non-Lambertian cases [64]–[66], while intrinsic decomposition methods exhibit a lack of robustness when applied to prominent Non-Lambertian surfaces [67], [68]. Moreover, few of these methods are designed for 3D scenes. We propose a method for detecting prominent Non-Lambertian surfaces based on uncertainty-driven detection. The approach directly optimizes Gaussian attributes using a simple regression loss, yielding masks of the prominent Non-Lambertian regions.

To model the distribution of Lambertian surfaces in the 3D scenes, we use the Lambertian attribute lb of the basic Gaussian to represent the likelihood of a surface being Lambertian. The lb is activated by a sigmoid function and initialized to 0.999. The rendered image $L(\mathcal{G}, T_{CW})$ that renders the lb of Gaussians $\mathcal{G}$ from camera pose T_{CW} is supervised by the conflict between the rendered image and the ground truth. Its photometric reconstruction loss is defined as follows:

$$L_{rgb} = E \left[\|I(\mathcal{G}, T_{CW}) - \bar{I}\|_2^2 \odot L(\mathcal{G}, T_{CW}) \right]. \quad (3)$$

Its depth reconstruction loss is defined as follows:

$$L_{depth} = E \left[\|D(\mathcal{G}, T_{CW}) - \bar{D}\|_2^2 \odot L(\mathcal{G}, T_{CW}) \right]. \quad (4)$$

A regularization loss is applied to all Gaussian lb attributes, defined as:

$$F_{lr} = E [|lb - lb^*|], \quad (5)$$

where the target value is $lb^* = 1$. The overall loss function is defined as:

$$L_{lb} = \lambda_1 L_{rgb} + \lambda_2 L_{depth} + F_{lr}. \quad (6)$$

Prominent Non-Lambertian surfaces exhibit stronger color anisotropy, which is difficult to capture using low-order spherical harmonics. As a result, the rendered reconstruction in these regions typically incurs higher losses, both in geometry and color. This essentially reveals the multi-view inconsistency of the reconstruction. We exploit this property by detecting prominent Non-Lambertian surfaces based on the conflict between the rendered colors and depths with their ground-truth across multiple views. As shown in Equation (3), the squared term of the per-pixel photometric reconstruction loss is multiplied by the corresponding value of $L(\mathcal{G}, T_{CW})$. Under the penalty in Equation (5), higher photometric reconstruction losses drive the corresponding $L(\mathcal{G}, T_{CW})$ toward smaller values. The use of the squared term encourages polarization of the photometric reconstruction loss, resulting in faster separation. L_{depth} is applied the same strategy. The optimization iterates over all keyframes to apply the updates, with all attributes being frozen but the lb being optimized. The pose of current frame is obtained from the initial optimization in tracking, while the poses of other keyframes are obtained from a further refinement in the local window.

The values of $L(\mathcal{G}, T_{CW})$ are first normalized to $[0, 255]$ and binarized using a threshold τ_b . To suppress noise and smooth the edges, the binary image is subsequently processed with a median filter followed by an opening operation. Connected component analysis is then performed, retaining only regions with an area exceeding τ_c , resulting in the final binary mask. Pixels with a value of 0 in the final mask are assigned true, and all others false, yielding the image-space mask of prominent Non-Lambertian surfaces M_{nl} . The above pro-

cedure robustly identifies prominent Non-Lambertian regions while suppressing noise.

C. Efficient Mapping with NL-ESR

We build up an efficient mapping with NL-ESR for heterogeneous scenes containing prominent Non-Lambertian surfaces, which is constructed using both basic and reflective Gaussians. Specifically, we design a Gaussian splatting reconstruction neural network, i.e., GaussRecon, to improve the efficiency of estimating basic Gaussian parameters, while the joint optimization of both Gaussians enhances the reconstruction fidelity of prominent Non-Lambertian surfaces.

1) *GaussRecon*: To enhance the efficiency of our system, we design GaussRecon which extracts features from 9 frames of images with camera poses and efficiently estimates the parameters of the basic Gaussian through a single forward pass. The frames are the nearest neighbor keyframes in the mapping, and their poses are obtained either from initial optimization in tracking or a further refinement in the local window. As shown in Figure 3, we follow NeuralRecon [59] to construct three local sparse feature volumes at different levels from multi-view images within the sliding window, which are then fused into global feature volumes for incremental reconstruction. In contrast, the global feature volumes are used only for occupancy and Gaussian parameter prediction, while TSDF prediction is not performed. Specifically, based on the spatial positions of the basic Gaussian, we query the 3 nearest neighboring volumes in the global feature volumes, and compute the average of the features to obtain the f_{im} . Meanwhile, the parameters of the Gaussian, including position, color, scale, rotation, and opacity, are concatenated to form the feature f_{gs} . Subsequently, the concatenated f_{im} and f_{gs} are fed into a 3-layer MLP to predict the offset of the Gaussian. The output is then added to the corresponding parameters of the original Gaussian, yielding the final predicted Gaussian.

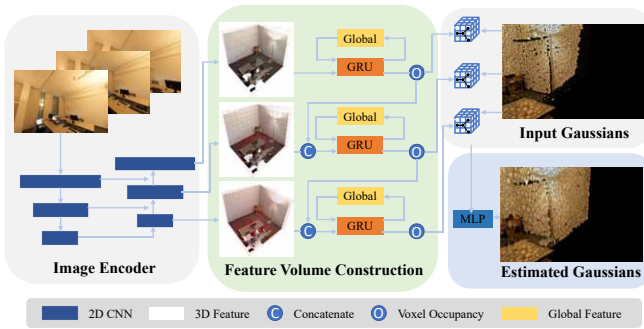


Fig. 3. The pipeline of GaussRecon. We first employ an image encoder to extract features at 3 scales, which are then back-projected into corresponding feature volumes. The global features across all scales are fused using a GRU network, and the resulting feature is fed into MLP to predict 3D Gaussian parameters.

The voxel occupancy loss L_{occ} is computed using Binary Cross-Entropy, where the ground truth is generated from the depth map. In contrast, the final predicted Gaussians are rendered based on the intrinsic and extrinsic parameters of the 9 frames of images, and their photometric reconstruction

loss L_{rebi} is computed. Additionally, we apply an isotropic penalty term on the scale of the 3D Gaussians, and its loss function is defined as follows:

$$F_{iso} = E \left[\sum_{i=1}^{|G|} \|s_{3d} - \overline{s_{3d}}\|_1 \right], \quad (7)$$

where s_{3d} denotes the 3 scales of a 3D Gaussian, $\overline{s_{3d}}$ represents the set of mean values of the 3 scales for each 3D Gaussian. The final loss function is as follows:

$$L_{nn} = L_{occ} + \lambda_3 \sum_{i=1}^9 L_{rebi} + \lambda_4 F_{iso}. \quad (8)$$

The above loss is applied at 3 scale levels, and optimization is performed after summing the losses across all levels.

The GaussRecon is pretrained on ScanNetV2 [69] and ScanNet++ [29] dataset, and is utilized to efficiently provide geometric priors for our NLGS-SLAM in an online manner. Specifically, once the current frame is selected as a keyframe, the new basic Gaussians are initialized using the pose, depth, and color of the keyframe, following a strategy similar to MonoGS [19]. Due to the lack of geometric priors, the new basic Gaussians initially struggle to capture the scene's features effectively. To address this, we input the images and camera poses of the 9 nearest keyframes, and all the basic Gaussians into GaussRecon to estimate 3D Gaussian parameters. It is worth noting that the new basic Gaussians input to GaussRecon are initialized using both color images and depth maps, rather than being generated directly from volume centers as in other pipelines. This estimation not only provides an absolute scale for the 3D Gaussian scene representation but also leverages the depth maps as strong priors to enhance reconstruction quality. These Gaussians are further fed into a local window for jointly refining basic Gaussians and the camera pose, guided by the photometric loss Equation (1) and geometric loss Equation (2). During this process, both the camera pose and the basic Gaussians, particularly those in Lambertian regions, are significantly optimized.

2) *NL-ESR Optimization*: After reconstructing keyframe containing Non-Lambertian surfaces with basic Gaussian, the mapping performs joint optimization of NL-ESR with basic and reflective Gaussians. In this stage, the initialization of reflective Gaussians is guided by the spatial distribution of the existing basic Gaussians. Specifically, we partition the scene's bounding box into 32^3 sub-grids and randomly sample 5 primitives within each grid to initialize the reflective Gaussian. The scene's bounding box is determined by the 95% quantile of the positions of the basic Gaussians. Moreover, the color of reflective Gaussian is initialized to 0, the opacity to 0.1, and the scale is determined by the distance between neighboring reflective Gaussians. It's worth noting that the initialization of reflective Gaussians is performed only once, after encountering the first keyframe containing prominent Non-Lambertian surface. During subsequent optimization, the reflective Gaussians are dynamically densified and pruned based on their attributes and gradients.

To balance reconstruction efficiency, ray tracing is performed only once in the reflective Gaussians. Specifically, the

depth and normal maps of the current image are obtained via rasterization of the basic Gaussians. Using M_{inl} , specific pixels requiring ray tracing are identified. These pixels in the depth map are then used as the starting points o for ray tracing, with the corresponding pixels in the normal map used to compute the specular reflection direction d_{ref} . The function for calculating d_{ref} is as follows:

$$d_{ref} = d_{cam} - 2(d_{cam} \cdot n)n, \quad (9)$$

where d_{cam} denotes the camera ray direction, and n represents the surface normal vector.

By representing each reflective Gaussian as two triangles, our ray tracing strategy is efficiently deployed on OptiX. Rays from o along d_{ref} are cast to intersect reflective Gaussians until no further intersections are found along the ray. The specular reflection image c_{ref} is obtained by ray tracing, with the remaining pixels filled with 0, while the basic rendering image is obtained by basic Gaussian rasterization. Integrating M_{inl} into the rendering process reduces the computation of start points, specular reflection directions, and ray tracing, leading to a significant enhancement in rendering efficiency. The final rendered image c is blended by c_{bas} and c_{ref} , weighted by a specular reflection intensity β , which is obtained by multiplying M_{inl} with the rendered result of $stren$. The function for calculating c is as follows:

$$c = (1 - \beta) \cdot c_{bas} + \beta \cdot c_{ref}. \quad (10)$$

To mitigate the interference of the forgetting problem of the basic Gaussian, joint optimization of the basic Gaussian and reflective Gaussian is performed across all keyframes. This process employs the photometric reconstruction loss, which combines L1 loss and a D-SSIM term with a ratio of 0.8 and 0.2, respectively.

IV. EVALUATION

We conduct a comprehensive evaluation of our system, comprising both qualitative visualizations and quantitative comparisons on diverse real-world scenes featuring mirrors, glass, and highlights. Subsequently, the effectiveness of each system component is demonstrated through module analyses and ablation study.

A. Experimental Setup

1) *Datasets*: We primarily evaluate our system on the ScanNet++ dataset [29], which provides more accurate depth and color information for high-fidelity reconstruction. Reconstructing scenes with prominent Non-Lambertian surfaces is more challenging than in previous works, and ScanNet++ [29] effectively highlights the system's reconstruction capabilities. 8 sequences are selected from ScanNet++ [29], each containing objects with one or more prominent specular reflections, such as mirrors, windows, and paintings. To further evaluate the robustness of our approach, experiments are also conducted on TUM dataset [30] following MonoGS [19].

2) *Implementation Details*: We run our SLAM on a computer with Intel Core i9 12900F 2.4GHz and a single NVIDIA GeForce RTX 4090. To accelerate reconstruction, prominent Non-Lambertian reconstruction is performed every 15 keyframes containing prominent Non-Lambertian surfaces. For the loss function, we set $\lambda_1 = 0.2$, $\lambda_2 = 0.8$, $\lambda_3 = 0.05$, and $\lambda_4 = 0.5$. For segmentation, we set $\tau_b = 30$ and $\tau_c = 1000$. The GaussRecon is trained on 136 sequences from ScanNetV2 dataset [69] and 17 sequences from ScanNet++ dataset [29], with no overlap with the sequences used for evaluation. Training is performed using the pre-trained weights of NeuralRecon [59] to accelerate convergence. For the 0c, 6f, and b2 sequences of the ScanNet++ dataset [29], M_{inl} is obtained using uncertainty-driven prominent Non-Lambertian segmentation, while the remaining sequences rely on manual annotations. During the optimization process, we also apply the normal propagation and color sabotage strategies, similar to [26]. Specifically, to enhance the global consistency between the Lambertian attribute of the basic Gaussian and the M_{inl} , we supervise an additional attribute of the basic Gaussian using the mask. Gaussian with this attribute falling below a specified threshold is considered to represent prominent Non-Lambertian surfaces, to which special strategies, such as normal propagation, are applied. During the inference stage, the rasterization result of this attribute is also used to generate the mask of Non-Lambertian surface region. Due to the large inter-frame spacing in the 1a, 0c, and b2 sequences, nearly all frames participate in reconstruction; evaluation is therefore performed only on the trained viewpoints, while novel view synthesis is assessed for the other sequences. For the TUM dataset [30], which contains few prominent Non-Lambertian regions, segmentation is not applied, and ray tracing is performed across all pixels.

3) *Metrics*: Camera tracking is evaluated using the Root Mean Square Error (RMSE) of the Absolute Trajectory Error (ATE) on keyframes, while reconstruction quality is assessed with standard photometric metrics, including Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), and perceptual similarity (LPIPS), following the evaluation protocol of MonoGS [19]. PSNR in dB is reported.

4) *Baseline Methods*: As the 8 sequences selected from ScanNet++ dataset [29] have been rarely used in prior work, we compare with state-of-the-art RGB-D SLAM methods, including MonoGS [19], 4DGS-SLAM [25] and SEGGS-SLAM [21], using publicly available implementations. Furthermore, extensive comparisons are conducted on TUM dataset [30] with existing advanced RGB-D SLAM methods, such as NICE-SLAM [14], ESLAM [16], Point-SLAM [17], CG-SLAM [20], and VarSplat [70]. The best results are marked as **best score**, **second best score** and **third best score**.

B. Quantitative Evaluation

1) *Camera Tracking Accuracy*: Table I reports the tracking results of our method on ScanNet++ dataset [29], together with comparisons to state-of-the-art methods. Our method achieves robust and accurate tracking across 8 scenes, with the best average performance as well as the best results on the 1a, 3b, 0c, 6f and b2 sequences.

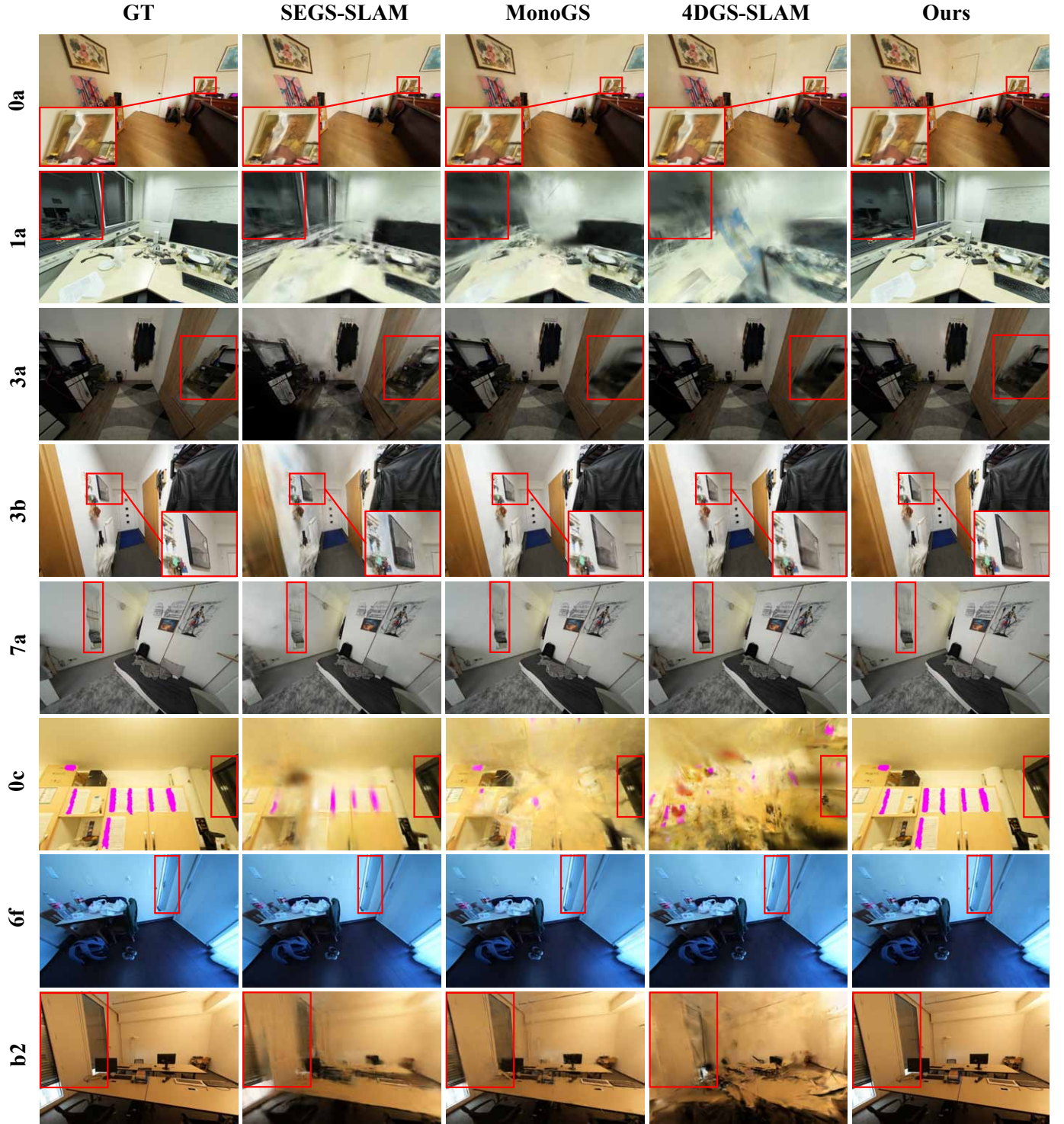


Fig. 4. Some qualitative comparison of rendering results between our NLGS-SLAM and other previous methods on ScanNet++ dataset.

Table II reports the tracking results of our method on TUM dataset [30], together with comparisons to state-of-the-art methods. Although our method is not tailored for datasets such as TUM dataset [30], where low-precision reconstructions make reliable ray tracing challenging, it still achieves performance comparable to the best existing methods and attains the highest tracking accuracy on fr1/desk. These results

demonstrate strong accuracy and generalization capability of our method.

2) *Rendering Performance*: Table III reports the rendering performance of our method on ScanNet++ dataset [29], together with comparisons to state-of-the-art methods. Our method achieves the best average performance across 8 sequences, with the largest number of best results on PSNR,

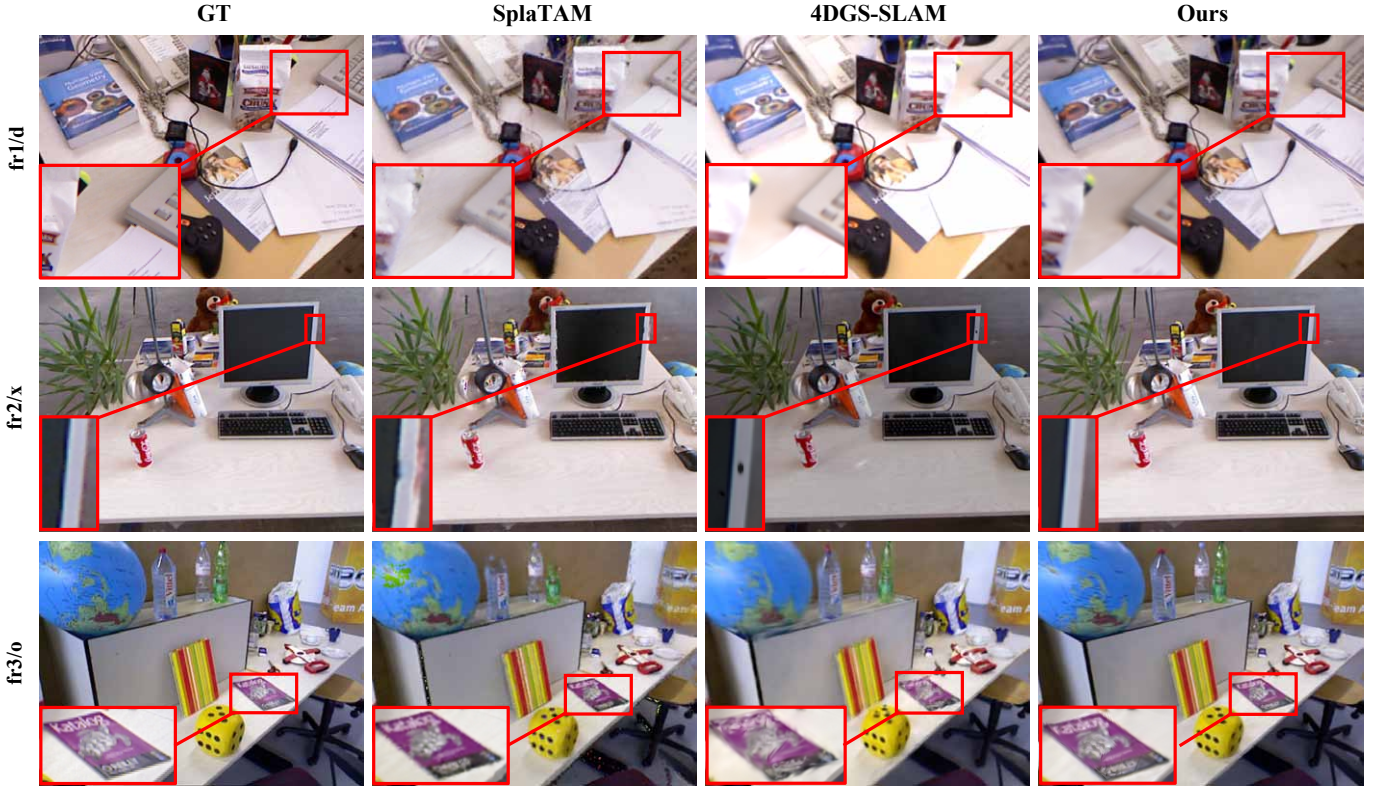


Fig. 5. Some qualitative comparison of rendering results between our NLGS-SLAM and other previous methods on TUM dataset.

TABLE I
CAMERA TRACKING RESULTS ON SCANNET++. ATE RMSE IN CM IS REPORTED.

Method	0a	1a	3a	3b	7a	0c	6f	b2	Avg.
MonoGS [19]	4.87	77.47	0.51	5.70	0.20	97.47	7.37	20.90	26.81
4DGS-SLAM [25]	1.38	89.18	0.58	5.81	0.20	99.42	0.78	40.01	29.67
SEGS-SLAM [21]	0.83	34.49	1.12	1.01	1.02	103.05	0.87	49.51	23.99
Ours	1.49	0.99	0.65	0.95	1.16	0.91	0.53	0.54	0.90

TABLE II
CAMERA TRACKING RESULTS ON TUM DATASET. ATE RMSE IN CM IS REPORTED.

Method	fr1/d	fr2/x	fr3/o	Avg.
iMAP [31]	4.90	2.00	5.80	4.23
NICE-SLAM [14]	2.70	1.80	3.00	2.50
ESLAM [16]	2.47	1.11	2.42	2.00
Co-SLAM [35]	2.40	1.70	2.40	2.17
Point-SLAM [17]	4.34	1.31	3.48	3.04
Loopy-SLAM [36]	3.79	1.62	3.41	2.94
MonoGS [19]	1.50	1.44	1.49	1.48
SplaTAM [7]	3.35	1.24	5.16	3.25
CG-SLAM [20]	2.43	1.20	2.45	2.03
GS-ICP SLAM [24]	2.70	1.80	2.70	2.40
Kintinous [71]	3.70	2.90	3.00	3.20
4DGS-SLAM [25]	1.57	1.51	1.73	1.60
VarSplat [70]	1.80	1.41	3.36	2.19
Ours	1.39	1.44	1.68	1.50

SSIM, and LPIPS, and consistently ranks within the top 3 across all metrics. Specifically, MonoGS [19] performs well on the 3a and 7a sequences, likely benefiting from better tracking in these cases. SEGS-SLAM [21] demonstrates strong performance on the 3b and 6f sequences, likely due to its anchor-based Gaussian splatting mechanism, which generates

significantly denser neural Gaussians, enabling more detailed scene representations. Notably, our method does not conflict with the baseline design of SEGS-SLAM [21], indicating that combining the two approaches may further improve performance, which we leave for future work. Overall, our method achieves state-of-the-art rendering quality on ScanNet++ dataset [29] and demonstrates strong potential for extension.

Table IV reports the rendering performance of our method on TUM dataset [30], together with comparisons to state-of-the-art methods. Our method demonstrates superior rendering performance on the fr1/d and fr3/o sequences, as well as on the average across all sequences. This highlights the robustness and generalizability of our method across various scene.

3) *Timing Analysis*: To evaluate the processing time of our system, we measure the total runtime required to process all frames on the TUM fr2/xyz dataset [30] and the ScanNet++ 3b dataset [29]. In addition, we render all images in each scene sequentially, and compute the average rendering speed as the total rendering time divided by the number of frames. As shown in Table V, despite performing the computationally expensive ray-tracing operations, our rendering speed still

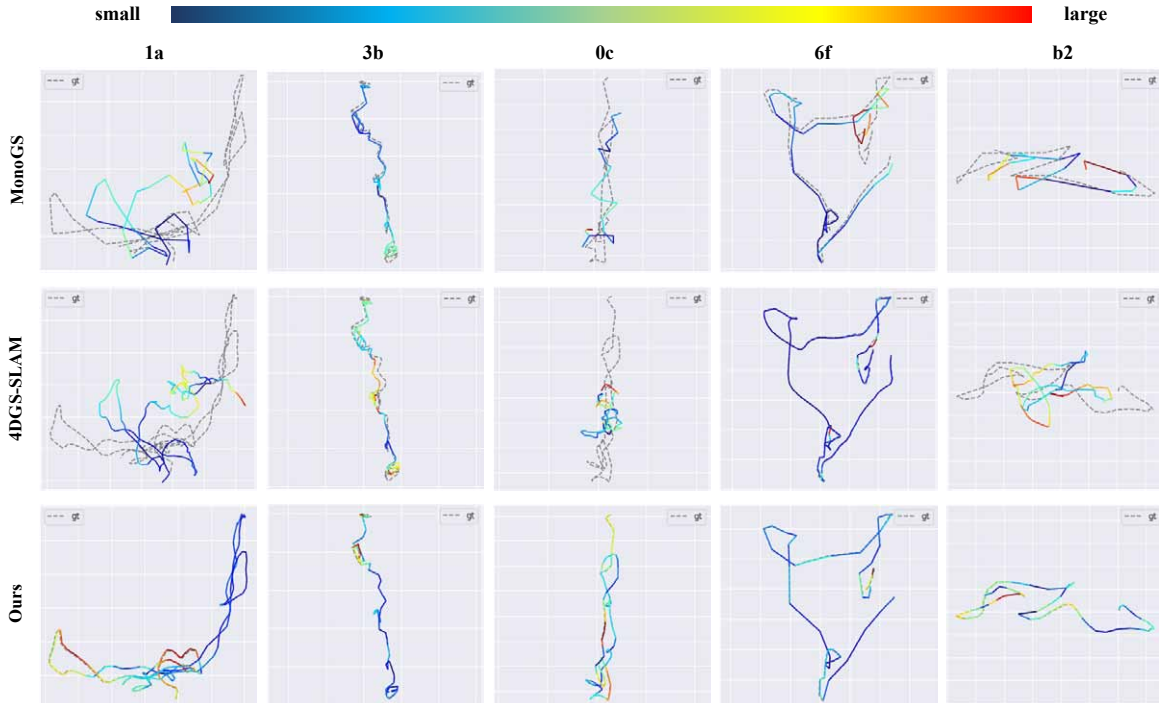


Fig. 6. Camera trajectory comparison on ScanNet++ dataset.

TABLE III
QUANTITATIVE RENDERING PERFORMANCE OF OUR METHOD COMPARED TO STATE-OF-THE-ART METHODS ON SCANNet++ DATASET.

Method	Metric	0a	1a	3a	3b	7a	0c	6f	b2	Avg.
MonoGS [19]	PSNR↑	29.65	22.92	34.66	27.68	34.07	24.34	29.54	27.59	28.81
	SSIM↑	0.927	0.822	0.948	0.903	0.957	0.866	0.902	0.895	0.903
	LPIPS↓	0.100	0.371	0.085	0.176	0.055	0.312	0.128	0.178	0.176
4DGS-SLAM [25]	PSNR↑	28.49	15.37	31.83	25.61	31.14	13.46	31.29	18.18	24.42
	SSIM↑	0.928	0.670	0.928	0.897	0.941	0.617	0.930	0.731	0.830
	LPIPS↓	0.118	0.564	0.113	0.213	0.082	0.667	0.102	0.488	0.293
SEGS-SLAM [21]	PSNR↑	28.51	28.02	29.81	29.96	29.48	23.82	32.93	31.44	29.25
	SSIM↑	0.931	0.873	0.900	0.934	0.911	0.821	0.935	0.918	0.903
	LPIPS↓	0.042	0.182	0.065	0.059	0.040	0.278	0.045	0.131	0.105
Ours	PSNR↑	30.30	30.86	32.83	29.84	32.29	31.69	29.91	35.96	31.71
	SSIM↑	0.941	0.931	0.934	0.937	0.943	0.939	0.904	0.963	0.937
	LPIPS↓	0.086	0.094	0.087	0.136	0.058	0.110	0.101	0.055	0.091

achieves 124.90 FPS.

4) *Ablative Analysis*: To evaluate the real-time rendering advantage of segmentation based prominent Non-Lambertian reconstruction, we conducted an ablation study on the ScanNet++ 3b dataset [29]. Specifically, instead of performing ray tracing only on prominent Non-Lambertian regions, we applied ray tracing to all pixels. As shown in Table VI, under comparable rendering performance, our method achieves a 4.07 times speedup, which is particularly beneficial for VR and AR applications that demand real-time performance.

To evaluate the effect of GaussRecon, an ablation study is conducted on ScanNet++ 3b dataset [29]. As shown in Table VII, increasing the mapping iterations of basic Gaussian from 200 to 500, without using GaussRecon enhances both camera tracking (ATE) and image rendering performance (PSNR) at the cost of a 76.03% increase in time consumption. In contrast, by using GaussRecon, camera tracking and image rendering can match those achieved with 500 mapping itera-

tions, even when the iterations are reduced to 200, resulting in a 68.28% improvement in computational efficiency. In addition, with 500 mapping iterations, the image rendering performance remains almost unchanged whether or not our GaussRecon is used. This indicates that our GaussRecon primarily improves mapping optimization efficiency, with minimal effect on the final optimization results.

C. Qualitative Results

1) *Qualitative Comparison*: We compare the rendering quality of our NLGS-SLAM with prior methods. As shown in Figure 4, SEGS-SLAM [21] exhibits poor rendering across multiple scenes, largely due to its difficulty in maintaining reliable tracking in diverse scenarios. The resulting inaccuracies in tracking lead to failures in scene reconstruction. Although SEGS-SLAM [21] achieves relatively stable tracking on the 3a sequence, its rendering quality remains limited, likely because the anchor-based Gaussian generation strategy lacks

TABLE IV
QUANTITATIVE RENDERING PERFORMANCE OF OUR METHOD COMPARED TO STATE-OF-THE-ART METHODS FOR TUM DATASET.

Method	Metric	fr1/d	fr2/x	fr3/o	Avg.
NICE-SLAM [14]	PSNR↑	13.83	17.87	12.89	14.86
	SSIM↑	0.569	0.718	0.554	0.614
	LPIPS↓	0.482	0.344	0.498	0.441
ESLAM [16]	PSNR↑	11.29	17.46	17.02	15.26
	SSIM↑	0.666	0.310	0.457	0.478
	LPIPS↓	0.358	0.698	0.652	0.569
Point-SLAM [17]	PSNR↑	13.87	17.56	18.43	16.62
	SSIM↑	0.627	0.708	0.754	0.696
	LPIPS↓	0.544	0.585	0.448	0.526
GS-ICP SLAM [24]	PSNR↑	-	-	-	20.72
	SSIM↑	-	-	-	0.768
	LPIPS↓	-	-	-	0.218
SpliTAM [7]	PSNR↑	18.36	21.27	18.80	19.48
	SSIM↑	0.676	0.779	0.690	0.715
	LPIPS↓	0.303	0.189	0.333	0.275
MonoGS [19]	PSNR↑	23.59	24.70	24.64	24.31
	SSIM↑	0.784	0.794	0.835	0.804
	LPIPS↓	0.242	0.218	0.212	0.224
4DGS-SLAM [25]	PSNR↑	19.89	16.01	20.81	18.90
	SSIM↑	0.752	0.732	0.807	0.764
	LPIPS↓	0.280	0.255	0.236	0.257
VarSplat [70]	PSNR↑	21.71	23.79	23.46	22.99
	SSIM↑	0.730	0.814	0.775	0.773
	LPIPS↓	0.324	0.189	0.250	0.254
Ours	PSNR↑	23.77	24.59	25.85	24.74
	SSIM↑	0.792	0.797	0.856	0.815
	LPIPS↓	0.225	0.194	0.184	0.201

TABLE V
TIMING ANALYSIS.

Dataset	System Time[s]	System FPS	Render FPS
ScanNet++	1477.31	0.25	124.90
TUM	3731.38	0.91	12.38

robustness. MonoGS [19] demonstrates comparatively better reconstructions in several cases; however, it produces overly blurred mirror regions and insufficiently accurate highlight renderings. 4DGS-SLAM [25] nearly fails on the 1a, 0c, and b2 sequences, with notably suboptimal rendering performance in prominent Non-Lambertian regions across other sequences. In contrast, our method delivers both robust and accurate reconstructions across all tested scenes, producing faithful renderings of highlights, glass, and mirrors. As shown in Figure 5, our method also demonstrates superior rendering performance on TUM dataset. Furthermore, we present the camera trajectory comparison on ScanNet++ dataset in Figure 6. Our method follows the ground truth camera trajectory (shown as dashed lines) more closely, demonstrating accurate and robust tracking. These results indicate that our method achieves state-of-the-art performance, offering significant practical benefits for immersive VR and AR applications.

2) *3D Gaussian Estimation Analysis*: As shown in Figure 7, the initialized Gaussians fail to effectively represent the scene, whereas the Gaussians estimated using GaussRecon can capture a coarse scene structure. Moreover, the Gaussian ellipsoids visualized on the right side of Figure 3 indicate our estimation exhibits good geometric consistency such as walls, further demonstrating the effectiveness of our method.

TABLE VI
ABLATION STUDY ABOUT SEGMENTATION.

Method	PSNR	SSIM	LPIPS	Render FPS
w/o Segmentation	29.92	0.937	0.136	30.72
Ours	29.84	0.937	0.136	124.90

TABLE VII
ABLATION STUDY ABOUT GAUSSRECON. ATE RMSE IN CM IS REPORTED.

Method	System Time[s]	ATE	PSNR
w/o GaussRecon, iters=200	1412.29	1.18	29.40
Ours, iters=200	1477.31	0.95	29.84
w/o GaussRecon, iters=500	2486.04	1.06	29.86
Ours, iters=500	2536.23	0.98	29.86

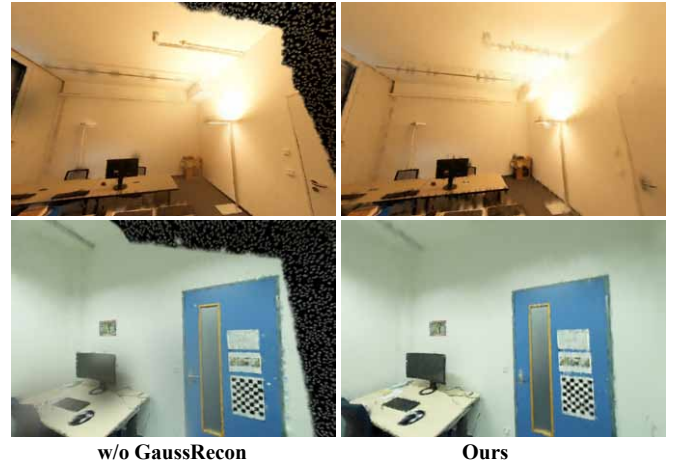


Fig. 7. 3D Gaussian estimation performance.

3) *Uncertainty-Driven Prominent Non-Lambertian Segmentation Analysis*: As shown in Figure 8, we render the color and lb of the basic 3D Gaussian. The results demonstrate that the basic 3D Gaussian not only captures the colors of glass and mirror regions but also embeds meaningful lb information, which can subsequently be leveraged to extract M_{int} .

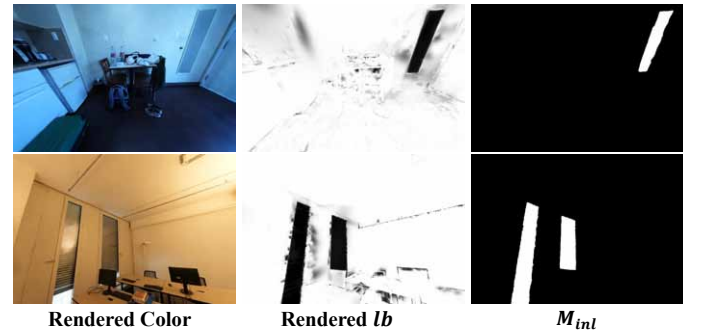


Fig. 8. Uncertainty-driven prominent Non-Lambertian segmentation analysis.

D. Limitation

Despite these advantages, our approach exhibits certain limitations. The frame-to-model incremental reconstruction strategy ensures higher consistency for Gaussians and poses, but inevitably incurs slower tracking. Additionally, existing

high-performing methods for prominent Non-Lambertian surfaces are generally computationally intensive. Due to these time-consuming operations, our method is better suited for scenarios emphasizing high-fidelity reconstruction rather than rapid scene reconstruction. The proposed prominent Non-Lambertian detection is particularly effective for objects with pronounced depth and color conflicts, such as mirrors and glass, but is less effective for objects with only subtle color conflicts, e.g., highlights on paintings. These improvements are left for future work.

V. CONCLUSIONS

We present NLGS-SLAM, a system integrates Lambertian property for robust tracking and Non-Lambertian enhanced representation, in which a Gaussian parameter estimation network is designed for efficient optimization, enabling high-fidelity reconstruction and real-time rendering of heterogeneous scenes in an online manner. Experimental results on public datasets demonstrate that NLGS-SLAM outperforms prior methods in reconstruction quality and significantly enhances the rendering fidelity of Non-Lambertian surfaces. With its high-quality reconstruction performance and real-time rendering, NLGS-SLAM supports highly immersive VR and AR experiences. We hope that this work will motivate further research toward more generalized scene reconstruction, expanding the application potential of 3DGS-based methods in VR and AR.

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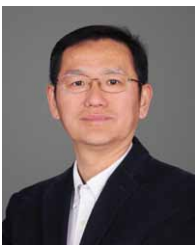
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Supplementary Material

I. IMPLEMENTATION DETAILS

For the scale of GaussRecon’s local fragments, the voxel size set to 6cm, and the number of voxels set to [144, 144, 144]. As shown in Figure 1, we present the method of estimating 3D Gaussian attributes from implicit features f_{im} . First, the f_{gs} is constructed by concatenating the parameters of the 3D Gaussian, including opacity, position, color, scale, and rotation, among them, opacity, scale, and rotation are represented in their unactivated states. Subsequently, f_{im} is concatenated with f_{gs} and passed through a 3-layer MLP decoder, with an output f_{os} , which is expected as the estimated offset of the Gaussian parameters. By adding the corresponding values of f_{gs} and f_{os} , the unactivated states of the 3D Gaussian parameters are obtained. Subsequently, the sigmoid function is applied for the opacity, the exponential function for the scale, the normalization function for the rotation, while no activation function is applied for the position and color parameters, producing the final 3D Gaussian parameters.

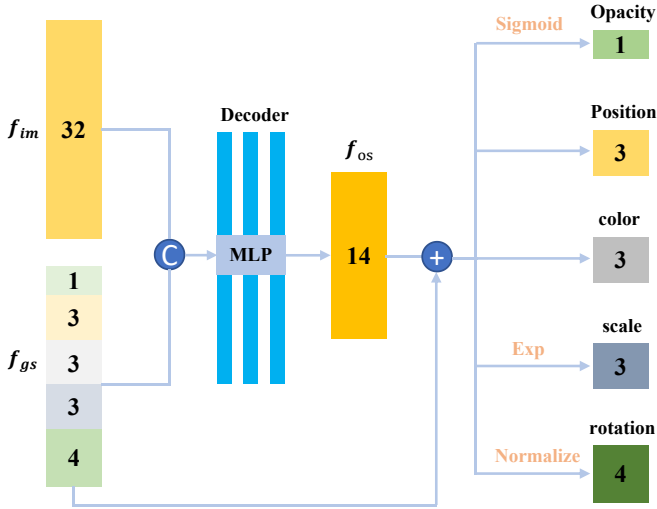


Fig. 1. Decoding 3D Gaussian parameters from implicit features f_{im} . After concatenating implicit features f_{im} and 3D Gaussian features f_{gs} , a 3-layer MLP decoder is applied to obtain f_{os} . And then, f_{os} is added to f_{gs} to derive the final parameters of 3D Gaussian.

II. ABLATIVE ANALYSIS

To evaluate the effect of the prominent Non-Lambertian mask M_{inl} , an ablation study is conducted on ScanNet++ dataset’s b2 sequence. As shown in Table I, when M_{inl} is not used in the photometric and depth losses, the tracking nearly fails, and the rendering performance is poor. This indicates that appropriately masking prominent Non-Lambertian regions in the loss may improve tracking robustness, thereby enhancing the scene reconstruction quality.

TABLE I

ABLATION STUDY ABOUT PROMINENT NON-LAMBERTIAN MASK. ATE RMSE IN CM IS REPORTED.

Method	PSNR[dB]	SSIM	LPIPS	ATE
w/o M_{inl}	27.56	0.907	0.167	45.12
Ours	35.96	0.963	0.055	0.54

To evaluate the effect of our reflective Gaussian with its ray tracing strategy, an ablation study is conducted on ScanNet++ dataset’s b2 sequence. As shown in Table II, the PSNR of the rendered image in both the prominent Non-Lambertian regions, i.e., NL, and the full image are enhanced when our reflective Gaussian, i.e., RG, with its ray tracing strategy is used. This demonstrates that the proposed representation NL-ESR, which combines both basic and reflective Gaussians, enhances the rendering quality of the prominent Non-Lambertian regions and the overall scene.

TABLE II

ABLATION STUDY ABOUT REFLECTIVE GAUSSIAN.

Method	NL PSNR[dB]	Full PSNR[dB]
w/o RG	29.69	34.85
Ours	32.03	35.96

III. MORE QUALITATIVE COMPARISON

To further demonstrate the rendering performance of our NLGS-SLAM, we present more qualitative comparison of rendering results between our NLGS-SLAM and other previous methods on ScanNet++ dataset in Figure 2, more qualitative rendering results of our NLGS-SLAM on ScanNet++ dataset in Figure 3, more qualitative comparison of rendering results between our NLGS-SLAM and other previous methods on TUM dataset in Figure 4, more qualitative rendering results of our NLGS-SLAM on TUM dataset in Figure 5. This demonstrates that our NLGS-SLAM can robustly reconstruct a diverse range of scenes, offering superior reconstruction quality compared to existing state-of-the-art methods. Furthermore, to demonstrate the geometric performance of our NLGS-SLAM, we present some qualitative comparison of rendering depth result between our NLGS-SLAM and other previous methods on ScanNet++ dataset in Figure 6. In addition, to further demonstrate the segmentation performance of our NLGS-SLAM, we present the image mask of prominent non-Lambertian surfaces generated during inference in Figure 7.

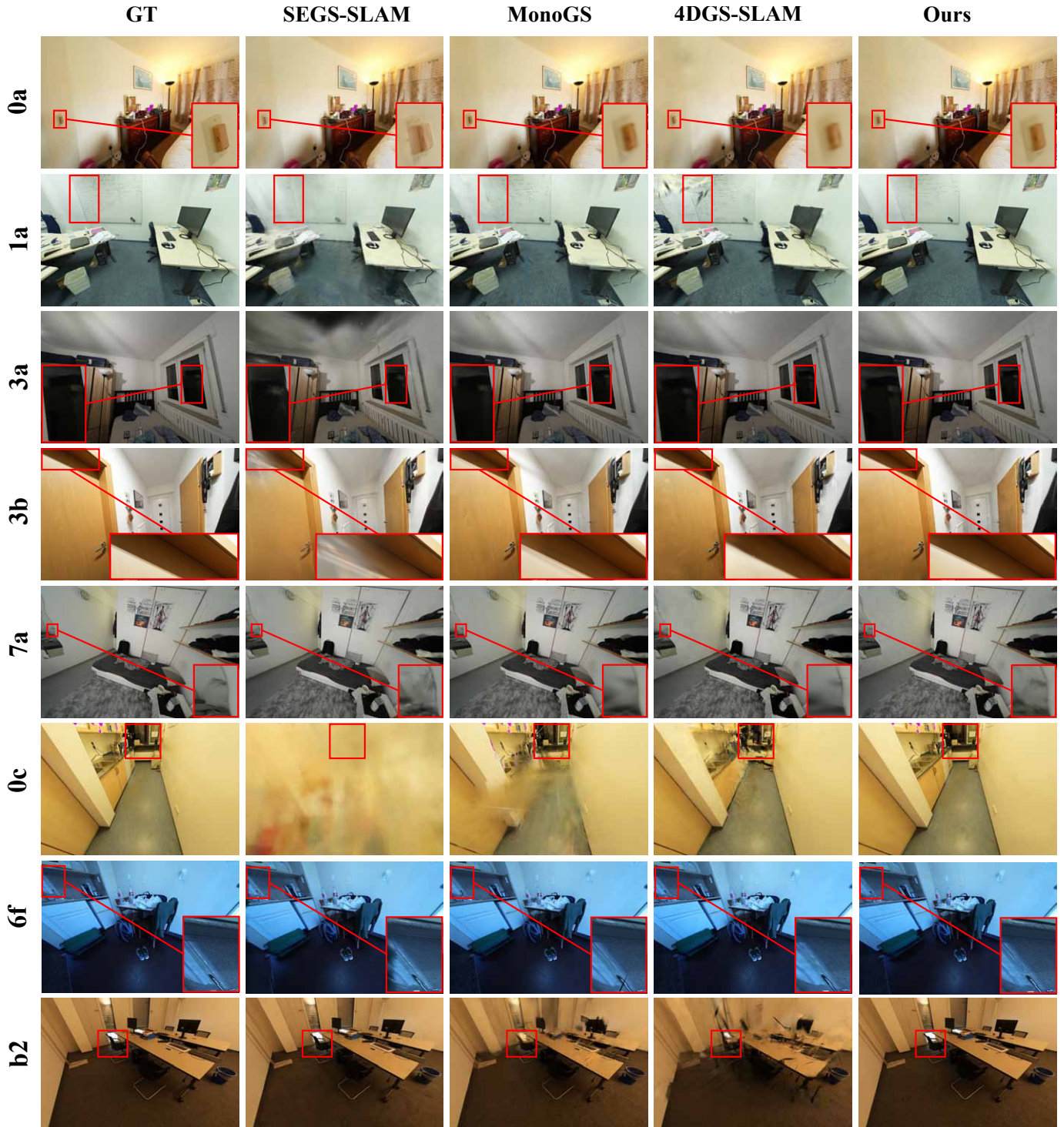


Fig. 2. More qualitative comparison of rendering results between our NLGS-SLAM and other previous methods on ScanNet++ dataset.

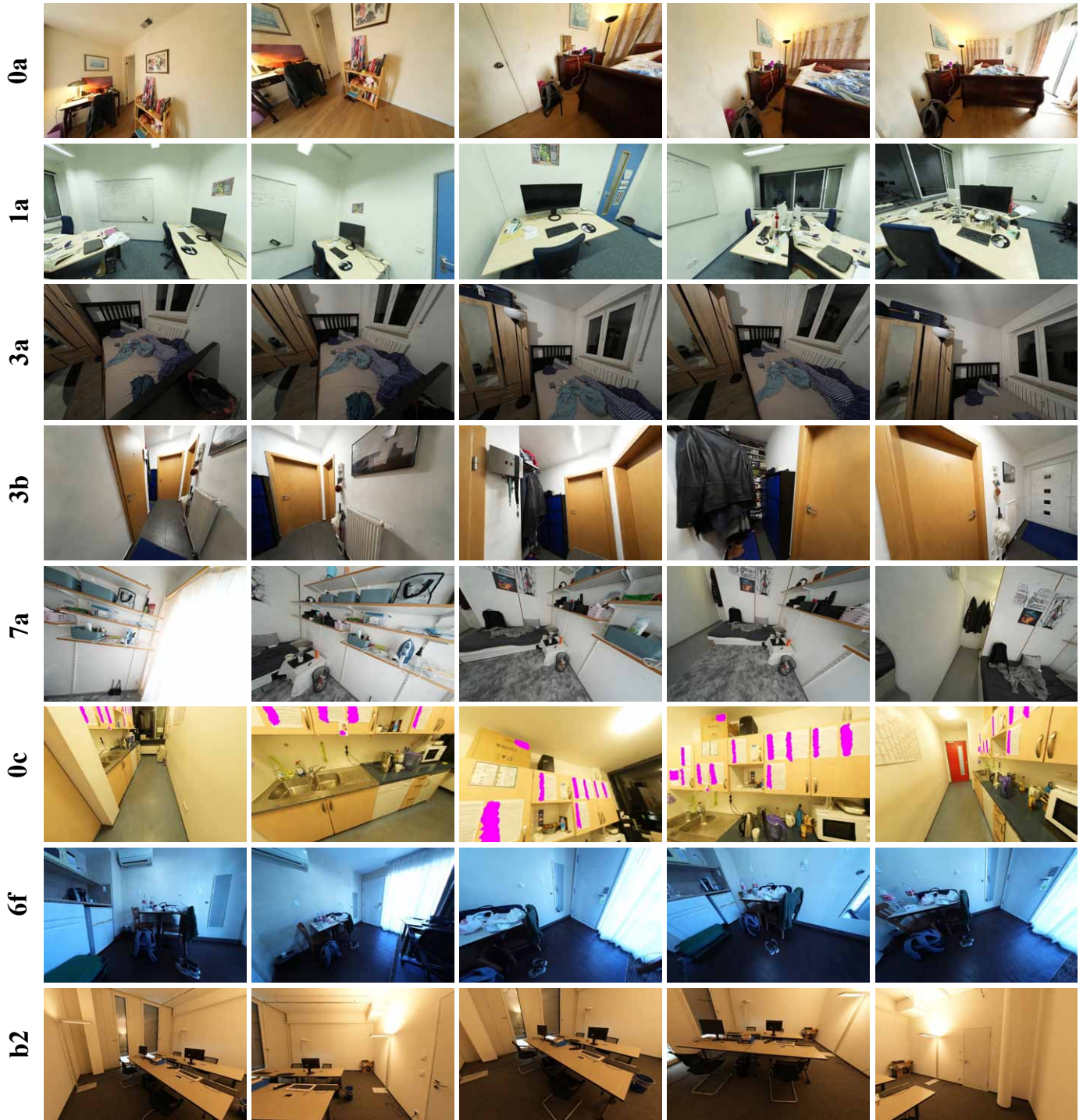


Fig. 3. More qualitative rendering result of our NLGS-SLAM on ScanNet++ dataset.

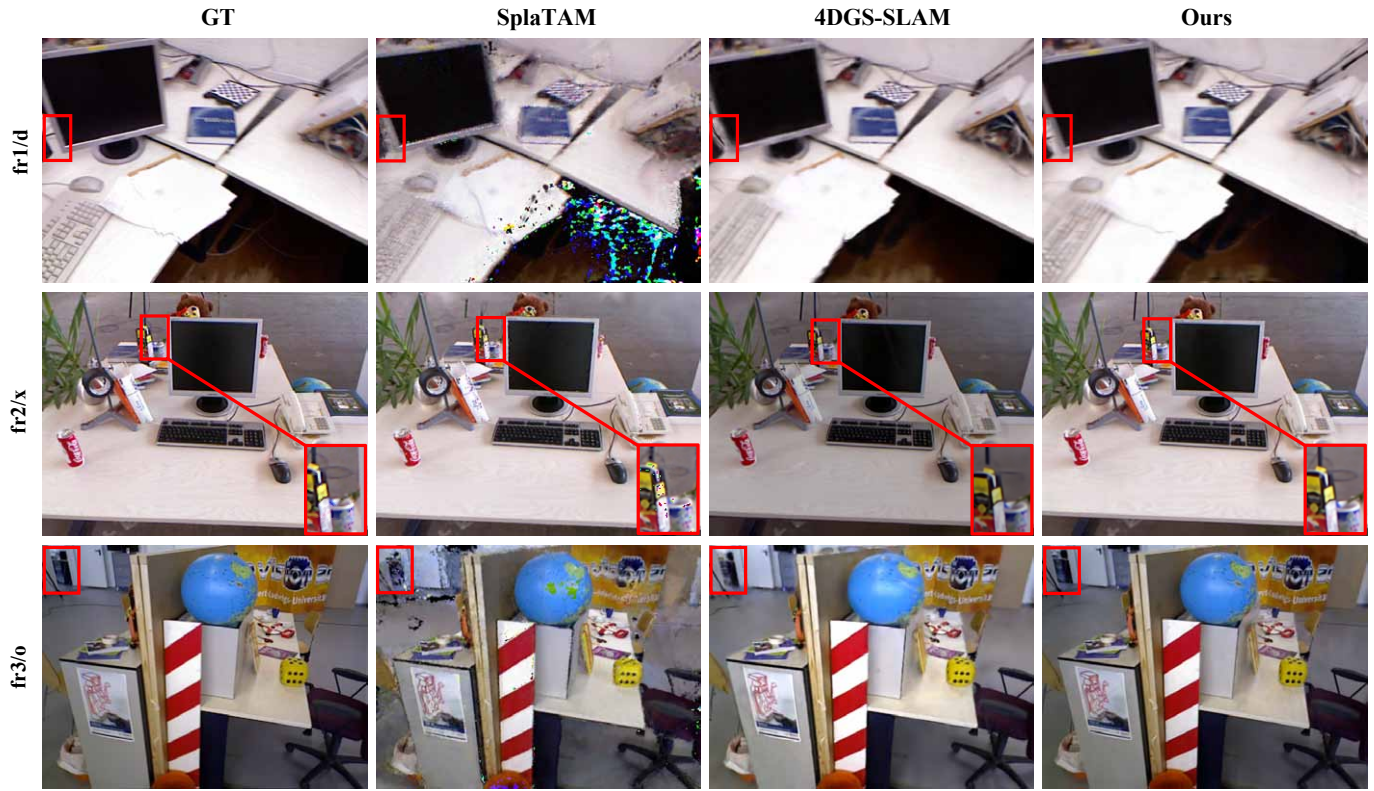


Fig. 4. More qualitative comparison of rendering results between our NLGS-SLAM and other previous methods on TUM dataset.

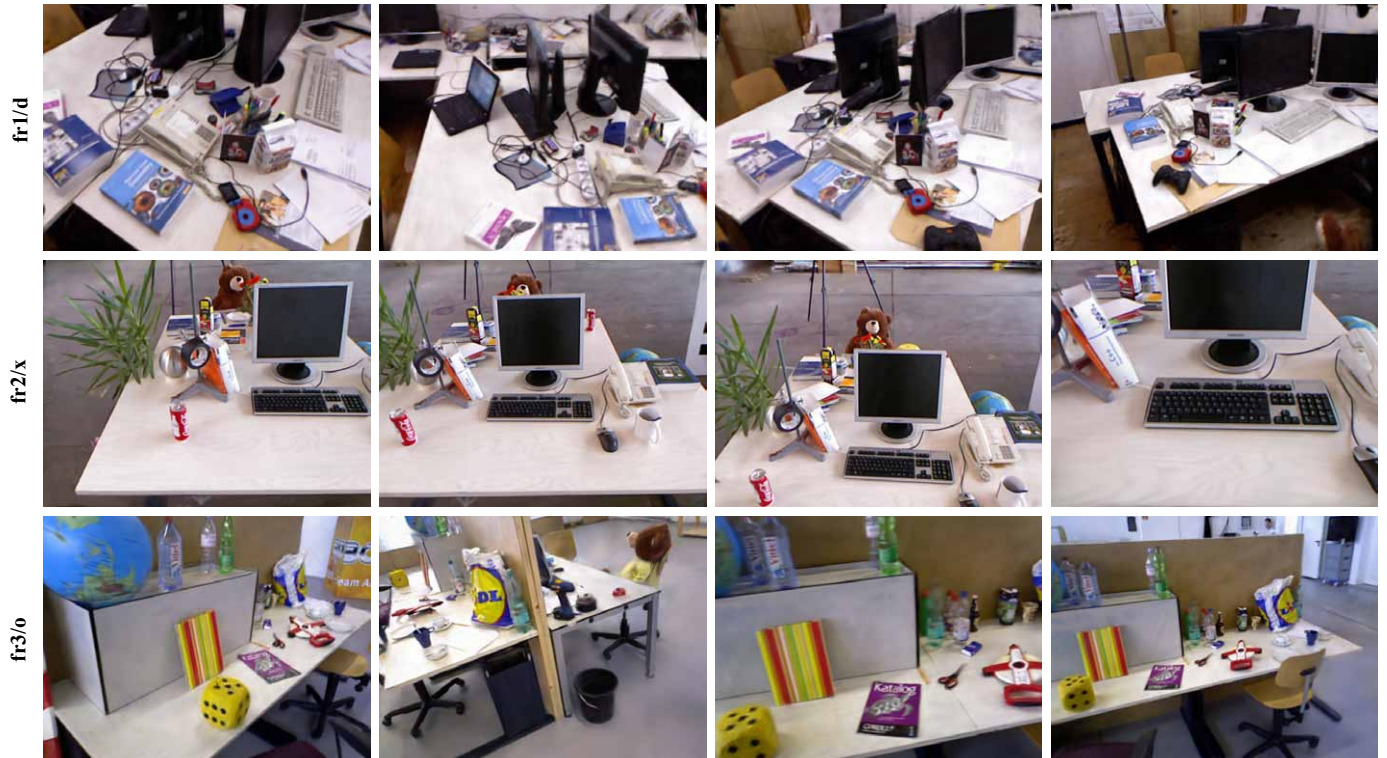


Fig. 5. More qualitative rendering results of our NLGS-SLAM on TUM dataset.

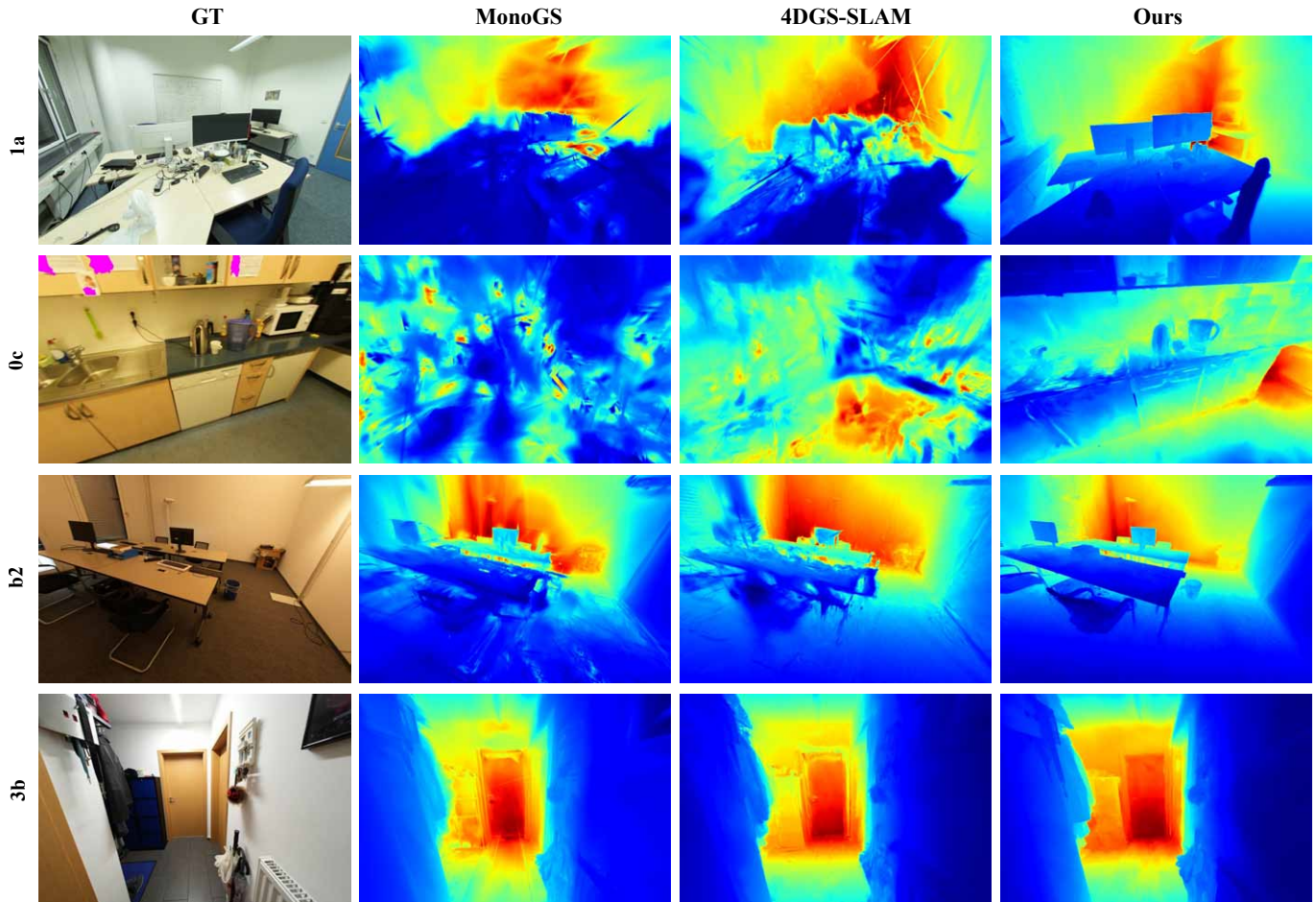


Fig. 6. Some qualitative comparison of rendering depth results between our NLGS-SLAM and other previous methods on ScanNet++ dataset.

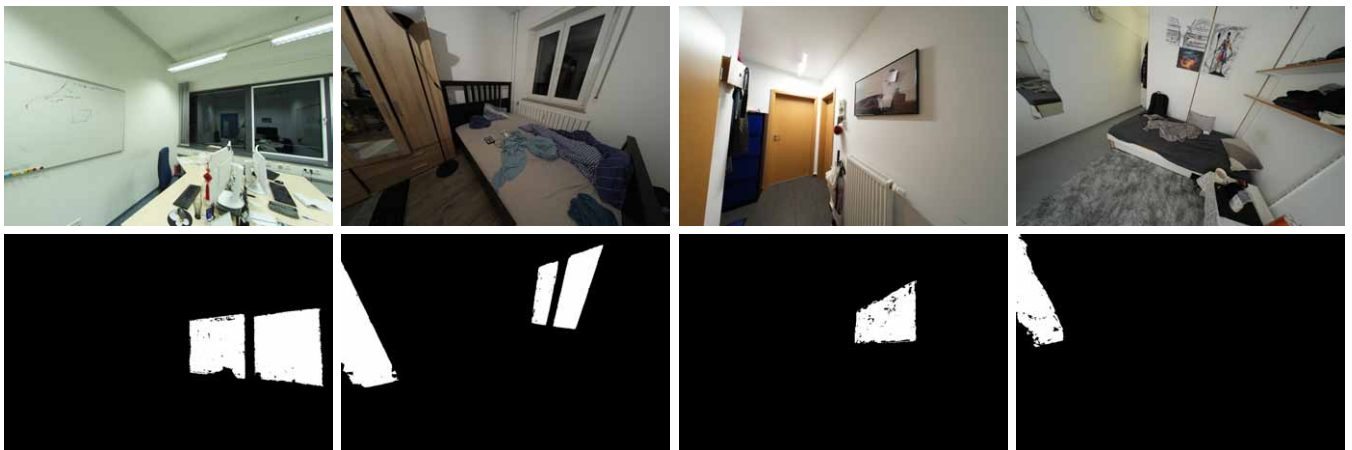


Fig. 7. The image mask of prominent non-Lambertian surfaces generated during inference. The first row is the original image, and the second row is the mask where the white regions represent prominent Non-Lambertian surfaces.